Non-Functional Test Cases: Performance Testing

Library Management System

Test Team (Coders & Non-Coders/Specification Experts)

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Note: These test cases are designed to evaluate the system's performance characteristics, including responsiveness, speed, and stability under various conditions. Metrics are illustrative and should be benchmarked against specific project requirements. Refer to the SRS and Test Plan for complete context.

IV.C Non-Functional Test Cases: Performance Testing

Table 1: Performance Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: UI Load & Responsiveness Testing			
TC_PERF_UI_001	Page load time for book list with few records	• **Precondition:** Database contains a small number of records (e.g., ¡ 50 books). • Login as a Librarian or Member. • Navigate to the book list page ('/books'). • Measure the time from navigation start to when the page is fully rendered and interactive.	• The page load time should be within the acceptable baseline (e.g., ¡ 2 seconds). • The UI is responsive and usable immediately after rendering.
TC_PERF_UI_002	Page load time for book list with many records (Volume)	• **Precondition:** Database contains a large number of records (e.g., ¿ 5,000 books). • Login as a Librarian or Member. • Navigate to the book list page ('/books'). • Measure the time until the page is fully rendered and interactive.	• The page load time meets the defined threshold for large data sets (e.g., ¡ 5 seconds). • UI remains responsive; features like sorting or filtering might show a loading indicator but do not freeze the application.
TC_PERF_UI_003	UI responsiveness during data entry on a high-load page	• Navigate to a data-heavy page (e.g., Librarian Dashboard with many widgets). • While the page is rendering or widgets are loading data, attempt to interact with a simple UI element (e.g., open a dropdown menu).	• User interactions are not blocked. • The UI responds to input within a reasonable time (e.g., ¡ 500ms) without noticeable lag.
Sub-Section: API Response Time Testing			
TC_PERF_API_001	API response time for fetching all books (Normal Load)	• **Precondition:** Database contains a moderate number of book records (e.g., 500). • Using an API testing tool (e.g., Postman), send a GET request to the '/api/book/getAll' endpoint. • Measure the server response time.	• The API response time is within the defined benchmark (e.g., ¡ 300ms). • The HTTP status code is 200 OK.

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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_PERF_API_002	API response time for creating a new entity (e.g., Borrowal)	• Using an API testing tool, send a POST request with a valid payload to the '/api/borrowal/add' endpoint. • Measure the server response time.	• The API response time for the creation operation is within the benchmark (e.g., ¡ 400ms). • The HTTP status code is 201 Created (or similar success code).
TC_PERF_API_003	API response time for user authentication	• Using an API testing tool, send a POST request with valid user credentials to the '/api/auth/login' endpoint. • Measure the server response time.	• Authentication is a critical path; response time should be fast (e.g., ¡ 250ms). • The HTTP status code is 200 OK.
Sub-Section: Volume Testing			
TC_PERF_VOL_001	System performance with large data volume in multiple tables	• **Precondition:** Populate the database with an abnormally large volume of data: – 20,000+ Books – 5,000+ Users – 50,000+ Borrowal records • Perform a series of common user actions (e.g., login, search for a book, view user list, view borrowal history).	• The system remains stable and does not crash or become unresponsive. • Data retrieval and display operations complete within defined acceptable limits, though slower than with few records. • Database queries do not time out.
Sub-Section: Stress Testing			
TC_PERF_STRS_001	System stability under high concurrent user load	• **Precondition:** Use a load testing tool (e.g., JMeter, Gatling). • Simulate a high number of concurrent users (e.g., 100 users) performing actions simultaneously for a sustained period (e.g., 15 minutes). Actions include: – Logging in – Fetching book lists – Adding new borrowals	• The system remains operational and does not crash. • The average API response time degradation is within an acceptable percentage (e.g., does not increase by more than 50 • The error rate (e.g., HTTP 5xx errors) is below a minimal threshold (e.g., ¡ 1
TC_PERF_STRS_002	System recovery after a stress period	• Following the completion of the TC_PERF_STRS_001 test, stop the load generation. • Wait for a brief period (e.g., 2 minutes). • Perform a single-user check by navigating key pages (Login, Books, Dashboard).	• The system recovers and returns to normal performance levels. • Response times for the single user should return to the baseline measurements recorded in the API and UI tests. • No data corruption has occurred.